

# TECHNICAL PORTFOLIO

## Vo Hong Quan

**AI Engineer: Reinforcement Learning for Robot Control**

Learning-based control for robots: modeled, trained and benchmarked in simulation.

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# Professional Profile

## Professional Summary

Robotics and AI engineer focused on learning-based control. Builds physical models, trains RL policies and benchmarks them against PID and LQR. intern role where learned control meets real machines.

## Core Competencies

|                                                                                                         |                                                                                                            |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>REINFORCEMENT LEARNING</b><br>PPO, SAC, reward design, hyperparameter tuning, frozen policy cascades | <b>CONTROL SYSTEMS</b><br>Cascaded PID, LQR, motor and sensor modeling, step-response analysis             |
| <b>ROBOTICS &amp; SIMULATION</b><br>NVIDIA Isaac Sim and Isaac Lab, PhysX, URDF from CAD, ROS 2         | <b>EMBEDDED DEPLOYMENT</b><br>Policy deployed on ESP32; networks sized for an STM32 flight controller      |
| <b>MACHINE LEARNING &amp; VISION</b><br>PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLO          | <b>TESTING &amp; EVALUATION</b><br>Randomized evaluation episodes, baselines, pytest, reproducible scripts |

## Tools & Technologies

- Design / simulation: NVIDIA Isaac Sim, Isaac Lab, PhysX, MATLAB (Simulink, Simscape, RL Toolbox), URDF
- Programming / firmware: Python, C/C++, MATLAB
- Hardware / platforms: modeled Crazyflie 2.1 Brushless, Pololu Balboa 32U4, Quanser QUBE-Servo 2; ESP32
- Management / collaboration: Git, GitHub, Conda

# Experience, Education & Achievements

## Experience / Internship / Research

|                    |                                                                                                                                                                                                                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feb 2025 – Present | <b>Research Assistant - Robotics Lab, Ho Chi Minh City University of Technology and Engineering</b> <ul style="list-style-type: none"><li>• Train and evaluate RL controllers for real-time robotic systems.</li><li>• Design reward functions and tune hyperparameters for stable policies.</li></ul> |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Education

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|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DEGREE</b><br>Bachelor of Robotics and Artificial Intelligence | <b>INSTITUTION</b><br>Ho Chi Minh City University of Technology and Engineering                                                                           |
| <b>TIME PERIOD</b><br>Aug 2022 – Dec 2026 (expected)              | <b>HIGHLIGHTS</b><br>GPA 3.22/4.0<br>Coursework: Machine Vision, Artificial Intelligence, Artificial Neural Networks, Practice of Artificial Intelligence |

## Certifications

- TOEIC Listening & Reading: 715/990 - IIG Vietnam - 2025

## Selected Project Portfolio

01

### Crazyflie-IsaacLab: RL-Tuned PID Gains for a Crazyflie Quadrotor

Small networks tune the PID gains of each layer of a four-layer flight cascade.

Individual project • Python, PyTorch, PPO, Isaac Lab • Jan 2026 – Sep 2026

Key result: 23–62% lower tracking error than the hand-tuned PID on all four layers.

02

### BalancingRobot-IsaacLab: Self-Balancing Robot with an RL Cascade

A velocity policy balances the robot; a frozen position policy drives it to a goal.

Individual project • Python, PPO, ROS 2, LQR, PID • Jun 2025 – Dec 2025

Key result: no falls in 768 randomized episodes; goals reached within 8 mm (median).

03

### RotaryPendulum-IsaacLab: Learned Cascade PID for a Rotary Inverted Pendulum

PPO retunes six gains of a two-loop PID at 100 Hz.

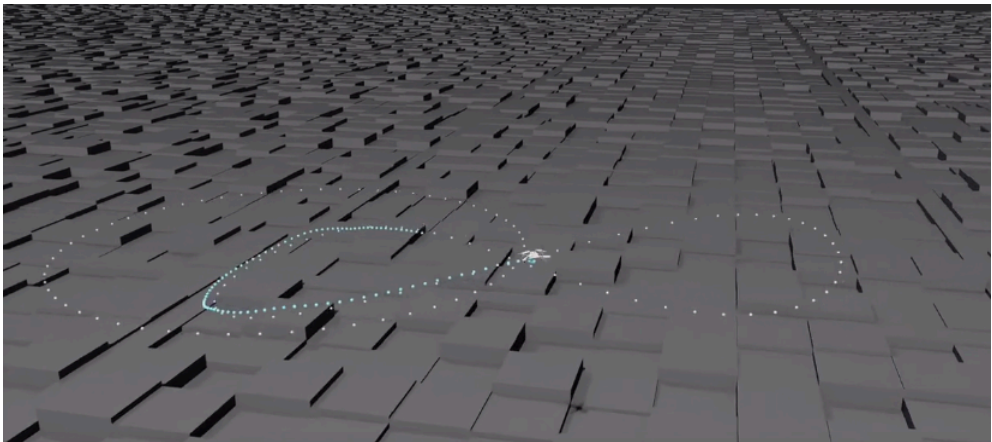
Individual project • Python, PPO (rsl\_rl), NVIDIA Isaac Lab, PhysX, URDF • Feb 2025 – May 2025

Key result: swing-up in 0.40 s across 256 robots with individual motor constants.

# Crazyflie-IsaacLab: RL-Tuned PID Gains for a Crazyflie Quadrotor

Small PPO networks tune each layer of a four-loop Crazyflie flight controller.

|                                                              |                                                                                                                                   |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>ROLE</b><br>Individual: modeling, training, evaluation    | <b>TIME PERIOD</b><br>01/2026 – 09/2026                                                                                           |
| <b>DOMAIN</b><br>Robotics / Control / Reinforcement learning | <b>LINKS</b><br>GitHub: <a href="https://github.com/vohongquann/Crazyflie-IsaacLab">github.com/vohongquann/Crazyflie-IsaacLab</a> |



|                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ENGINEERING PROBLEM</b><br>Track position through four loops at 50–500 Hz with networks small enough for the STM32. | <b>MY CONTRIBUTION</b> <ul style="list-style-type: none"> <li>• Built a four-layer flight controller (50–500 Hz) and physics-based Crazyflie model in Isaac Lab 3.0.</li> <li>• Trained one PPO MLP per layer to tune 9 PID gains, reducing tracking error 23–62% versus PID (attitude: 15.5° to 5.9°).</li> <li>• PID and RL gains layer by layer, wrote the report scripts, a pytest suite and a demo video.</li> </ul> |
|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Solution & System Architecture

|                                                                                    |
|------------------------------------------------------------------------------------|
| Commands: random targets, circle and figure eight                                  |
| Small PPO MLP: observations → 9 PID gains                                          |
| Cascade: position 50 Hz → velocity 100 Hz → attitude 250 Hz → rate 500 Hz → motors |
| Crazyflie model: motor curve, mixer, 1 kHz PhysX                                   |
| Evaluation: tracking error and motor chatter across 64 drones                      |

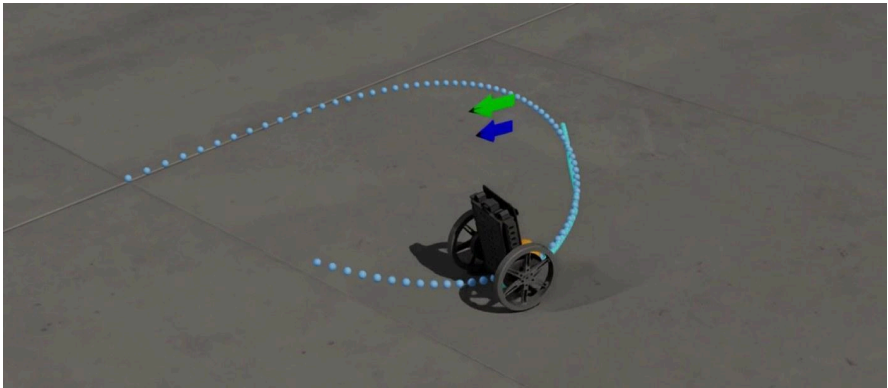
## Measured Results

|                                                       |                                                          |                                                             |
|-------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------|
| <b>15.5° → 5.9°</b><br>attitude error, PID → RL gains | <b>0.85 → 0.46 m/s</b><br>velocity error, PID → RL gains | <b>2.6–6.3×</b><br>motor chatter vs PID on the outer layers |
|-------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------|

# BalancingRobot-IsaacLab: Self-Balancing Robot with an RL Cascade

A velocity policy balances a Pololu Balboa; a frozen position policy drives it to a goal.

|                                                              |                                                                                                                                                |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ROLE</b><br>Individual: modeling, training, ROS 2         | <b>TIME PERIOD</b><br>06/2025 – 12/2025                                                                                                        |
| <b>DOMAIN</b><br>Robotics / Control / Reinforcement learning | <b>LINKS</b><br>GitHub:<br><a href="https://github.com/vohongquann/BalancingRobot-IsaacLab">github.com/vohongquann/BalancingRobot-IsaacLab</a> |



Position cascade following a figure eight.

|                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ENGINEERING PROBLEM</b><br>Balance and navigate from a noisy IMU and encoders under random starts and pushes. | <b>MY CONTRIBUTION</b> <ul style="list-style-type: none"><li>• Trained a PPO velocity policy that maps commands and raw IMU/encoder data to wheel torques.</li><li>• Added a frozen position policy; recorded zero falls in 768 randomized episodes and 8 mm median goal error.</li><li>• Beat LQR on speed steps: 0.026 m/s error in 0.36 s versus 0.067 m/s in 1.1 s.</li></ul> |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Solution & System Architecture

|                                                                   |
|-------------------------------------------------------------------|
| Noisy IMU and wheel encoders                                      |
| Complementary filter → position policy → frozen velocity policy   |
| Goal → speed/turn command → wheel torque; compare RL, LQR and PID |
| Pololu Balboa model in Isaac Lab / PhysX                          |
| 768 episodes: pushes, noise and randomized dynamics               |

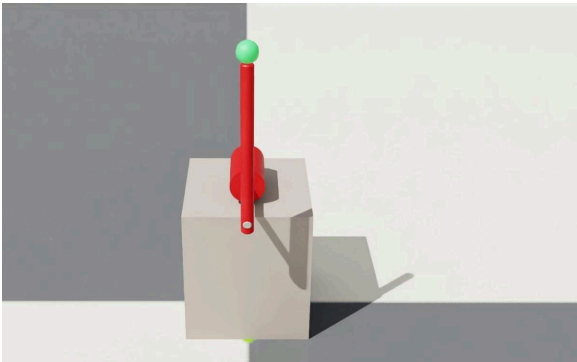
## Measured Results

|                                                            |                                                                     |                                                                                 |
|------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>0 falls</b><br>in 768 randomized episodes, every policy | <b>8 mm</b><br>median distance to the goal when the next goal comes | <b>0.026 m/s</b><br>speed-step error, settled in 0.36 s (LQR: 0.067 m/s, 1.1 s) |
|------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------|

# RotaryPendulum-IsaacLab: Learned Cascade PID for a Rotary Inverted Pendulum

PPO retunes both loops of a QUBE-Servo 2 cascade PID at 100 Hz.

|                                                              |                                                                                                                                                |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ROLE</b><br>Individual: modeling, training, evaluation    | <b>TIME PERIOD</b><br>02/2025 – 05/2025                                                                                                        |
| <b>DOMAIN</b><br>Robotics / Control / Reinforcement learning | <b>LINKS</b><br>GitHub:<br><a href="https://github.com/vohongquann/RotaryPendulum-IsaacLab">github.com/vohongquann/RotaryPendulum-IsaacLab</a> |



Pendulum balanced while tracking an arm-angle command.

|                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ENGINEERING PROBLEM</b><br>Swing up, balance and track arm angles with quantized encoders and varied motor constants. | <b>MY CONTRIBUTION</b> <ul style="list-style-type: none"><li>Modeled the QUBE-Servo 2 from CAD with a DC motor and quantized encoders at 500 Hz.</li><li>Trained a PPO policy on 4,096 robots in 64 minutes to tune six cascade-PID gains at 100 Hz.</li><li>Validated on 256 randomized robots: 0.40 s swing-up, 0.23–0.50 s settling and at most 1.4° median overshoot.</li></ul> |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Solution & System Architecture

|                                                                |
|----------------------------------------------------------------|
| Arm command, encoder angles and velocities (12 observations)   |
| PPO at 100 Hz: 12 observations → 6 PID gains                   |
| Outer PID 250 Hz → inner PID 500 Hz → voltage                  |
| QUBE model: PWM, current, back-EMF and torque; PhysX at 500 Hz |
| 256 robots with randomized motor constants                     |

## Measured Results

|                                                                |                                                     |                                                         |
|----------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------|
| <b>0.40 s</b><br>median swing-up time (95th percentile 0.48 s) | <b>0.23–0.50 s</b><br>median arm-step settling time | <b>≤ 1.4°</b><br>median overshoot, steady error ≤ 0.24° |
|----------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------|

# Technical Evidence & Contact

Clickable source code, results and videos.

## Repositories / Documents

|    |                         |                                                                                                                                                                |
|----|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | Crazyflie-IsaacLab      | Cascaded RL gain tuning, Crazyflie 2.1 Brushless:<br><a href="https://github.com/vohongquann/Crazyflie-IsaacLab">github.com/vohongquann/Crazyflie-IsaacLab</a> |
| 02 | BalancingRobot-IsaacLab | Frozen RL cascade for a Pololu Balboa:<br><a href="https://github.com/vohongquann/BalancingRobot-IsaacLab">github.com/vohongquann/BalancingRobot-IsaacLab</a>  |
| 03 | RotaryPendulum-IsaacLab | Learned cascade PID, QUBE-Servo 2:<br><a href="https://github.com/vohongquann/RotaryPendulum-IsaacLab">github.com/vohongquann/RotaryPendulum-IsaacLab</a>      |
| 04 | GitHub profile          | All repositories: <a href="https://github.com/vohongquann">github.com/vohongquann</a>                                                                          |
| 05 | Portfolio website       | Interactive demos and CV download: <a href="https://vohongquann.github.io/Portfolio">vohongquann.github.io/Portfolio</a>                                       |



GitHub profile: [github.com/vohongquann](https://github.com/vohongquann)

## Contact

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